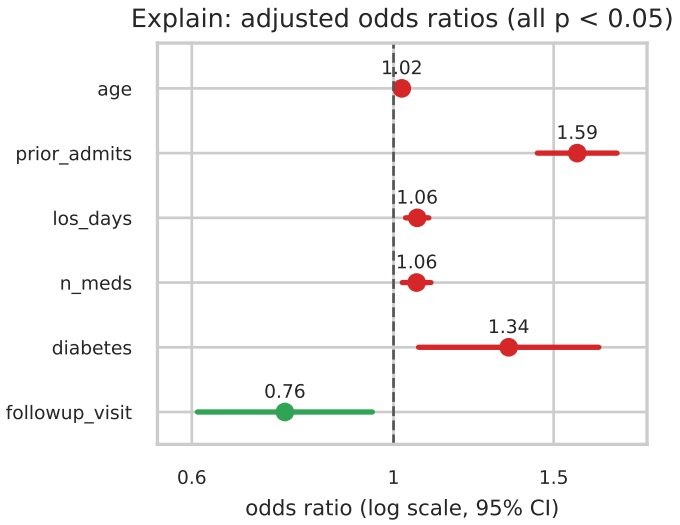
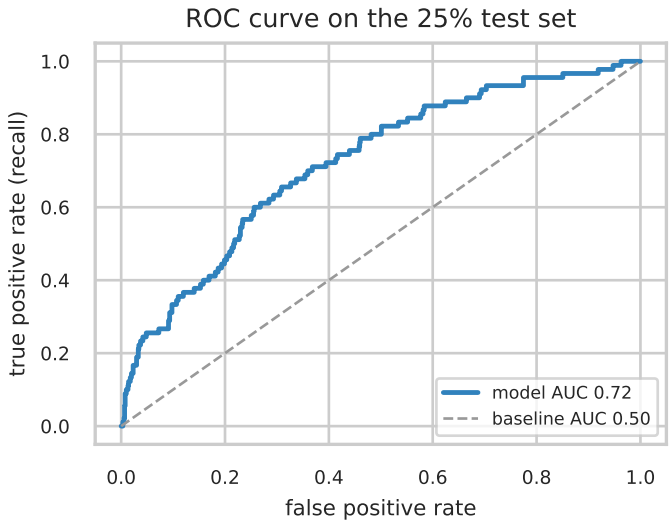
Day 14 Cheat Sheet: Explain vs Predict

Week 2 wrap-up: odds ratios answer 'why', a held-out test set answers 'who'

TERM	WHAT IT MEANS	CODE
Explain (infer)	which factors are linked to the outcome, how much	`smf.logit("y ~ x1 + x2", df).fit()`
Odds ratio (OR)	odds in group / reference, others held fixed	`np.exp(model.params)`
Predict	score new patients you have not seen yet	`LogisticRegression().fit(X, y)`
Train/test split	fit on one part, grade on a part kept hidden	`train_test_split(..., stratify=y)`
Baseline	dumbest honest model: everyone gets the avg rate	`np.full(n, y_train.mean())`
Probability	model's risk score between 0 and 1	`clf.predict_proba(X)[:, 1]`
AUC	chance a readmit is ranked above a non-readmit	`roc_auc_score(y, p)  (0.5 = coin)`
Brier score	mean squared error of the probabilities; lower=good	`brier_score_loss(y, p)`
Calibration	predicted 20% risk really means about 20% happen	`group by decile, compare pred vs obs`
Lift	rate in the top group / overall rate	`top.y.mean() / y.mean()`
Data leakage	info from the future or the test set in training	`split first, then fit anything`





THIS DATA (simulated, seed 14): 4,000 discharges, 6 features, 30-day readmission
Readmission rate 9.0% | train 3,000 / test 1,000 (stratified)
Odds ratios: prior admit 1.59 each, diabetes 1.34, 7-day visit 0.76, LOS 1.06 per day
Test AUC: model 0.723 vs baseline 0.500 | Brier: model 0.0771 vs baseline 0.0819
Top 10% of risk scores: 24% readmitted (lift 2.7x), catching 27% of all readmissions

HABITS THAT KEEP YOU HONEST
- Decide first: are you explaining (ORs, CIs) or predicting (test-set AUC, calibration)?
- Always beat a baseline, and always grade on data the model never saw
- A tiny p-value can sit next to a near-useless predictor (visit: p = 0.015, AUC 0.52)
- AUC ranks patients; calibration checks the risk numbers. Report both
- Translate to operations: "top 10% list holds about a quarter of readmissions"

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score, brier_score_loss
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=.25, stratify=y, random_state=14)
p = LogisticRegression(max_iter=1000).fit(Xtr, ytr).predict_proba(Xte)[:, 1]
```

Infographic: From 'Why' to 'Who' on 30-Day Readmissions

Simulated: 4,000 discharges. Week 2 statistics meets week 3 machine learning

Question	Split	Baseline	Fit	Grade	Act
explain or predict?	75% train 25% test	average rate AUC 0.50	logistic regression	AUC, Brier, calibration	top-decile outreach list

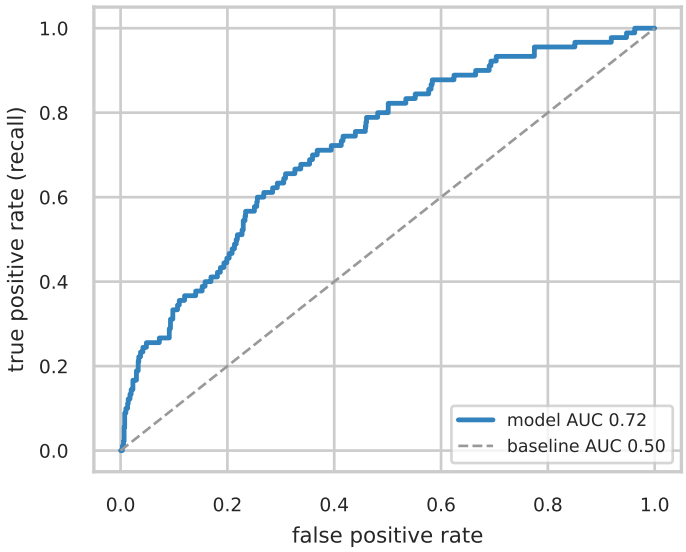
BASE RATE
9%
readmitted in 30d

TEST AUC
0.72
vs 0.50 baseline

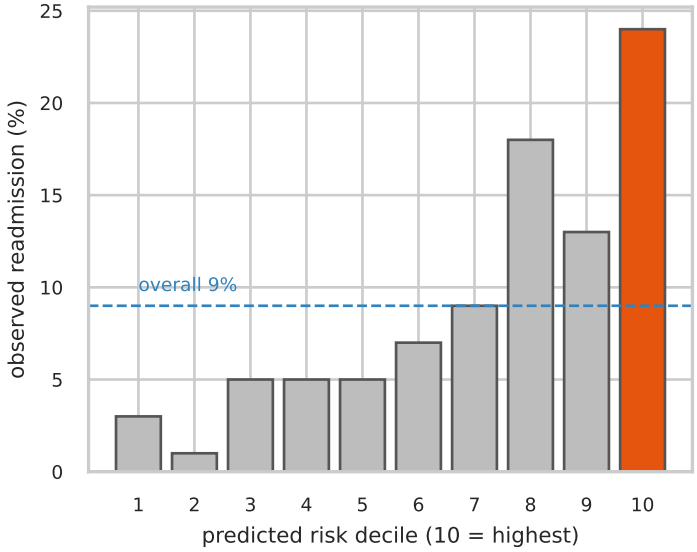
TOP DECILE
24% readmitted
lift 2.7x

PRIOR ADMITS
OR 1.59
strongest signal

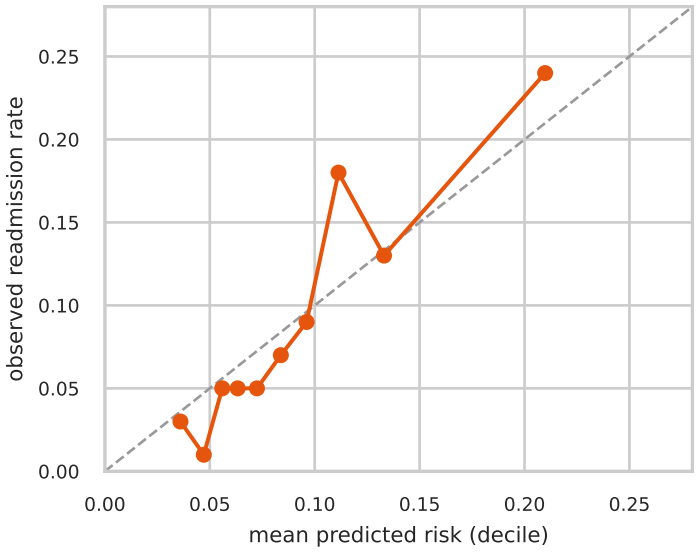
ROC curve on the 25% test set



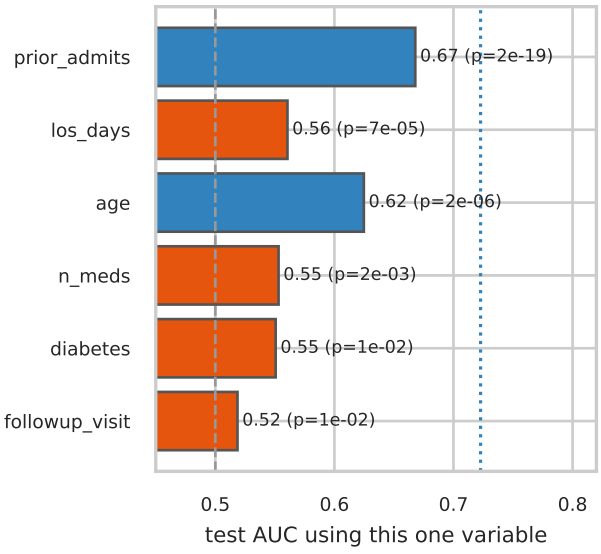
Top decile: 24% readmitted, lift 2.7x



Calibration: dots near the line = honest risks



Significant does not mean predictive



Read it like a care management lead: a simple model ranks patients well enough that the top 10% list holds 27% of readmissions at 2.7x the average rate. It is a starting point, not a deployed tool: the data are simulated, AUC 0.72 is moderate, and week 3 will try to beat it with better models.

Hands-On Worksheet: ds-genai-90-days / day14

60-90 minutes | notebook: day14/explain_vs_predict.ipynb + day14/README.md

[ ] mkdir day14, conda activate ds90, open explain_vs_predict.ipynb

[ ] Simulate readmissions_30d.csv with seed 14 (snippet 1)

[ ] Write the business question two ways: explain vs predict

[ ] Fit statsmodels logit; table of ORs, CIs, p-values (snippet 2)

[ ] Train/test split, stratified, random_state=14 (snippet 3)

[ ] Baseline: everyone gets the training average; record AUC, Brier

[ ] Fit LogisticRegression; record test AUC and Brier

[ ] Decile table: predicted vs observed, top-decile lift (snippet 4)

[ ] Plot ROC curve and calibration plot side by side

[ ] One-variable AUCs vs p-values (snippet 5): explain the gap

[ ] Week 2 README: question, data, 3 charts, 5 findings, limits

[ ] Link days 1-13 notebooks in the README as a week 2 index

[ ] git add day14 && git commit && git push (snippet 6)

[ ] Stretch: Snowflake SQL decile lift table (snippet 7)

```
import pandas as pd, numpy as np, statsmodels.formula.api as smf
df = pd.read_csv("readmissions_30d.csv")
m = smf.logit("readmit30 ~ age + prior_admits + los_days + n_meds + diabetes + followup_visit", df).fit()
print(np.exp(m.params))                                    # odds ratios
# then: split -> baseline -> LogisticRegression -> roc_auc_score, brier_score_loss
```

Rate: _____ OR prior_admits: _____ OR visit: _____ Baseline AUC: _____ Model AUC: _____

Brier base: _____ Brier model: _____ Top-decile rate: _____ Lift: _____ % caught: _____

Day 14 Self-Check Quiz

1. A CMO asks 'does the 7-day visit matter?' Explain or predict? Which output answers it?

_____

2. Why must the model be graded on a test set it never saw?

_____

3. What does an AUC of 0.72 mean in one plain sentence? What does 0.50 mean?

_____

4. The baseline and model Brier scores are close. Why is that normal at a 9% base rate?

_____

5. What is calibration, and why does a care team care about it, not just AUC?

_____

6. The 7-day visit has p = 0.015 but AUC alone of 0.52. How can both be true?

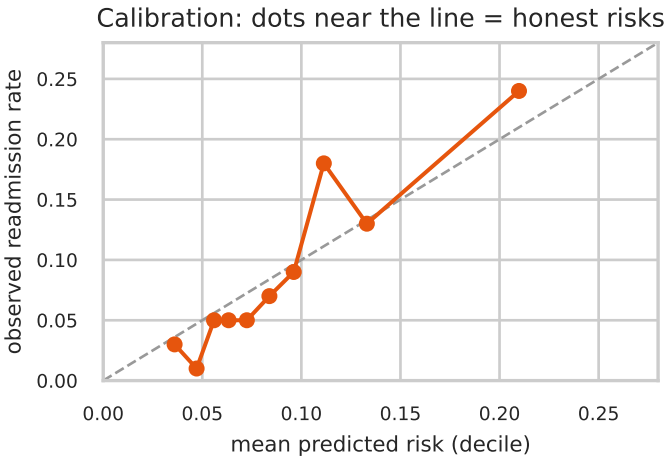
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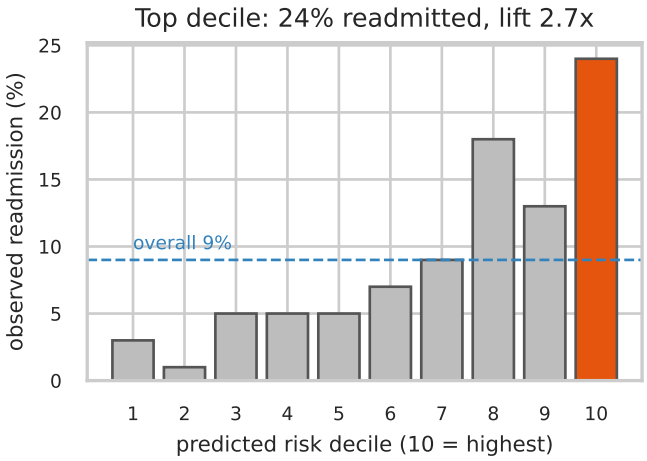
7. Top decile: 24% readmitted vs 9% overall. Say what this means for outreach.

_____

8. Name two things that would make this model unsafe to deploy as is.

_____





Answers:
1 explain: the logistic regression OR for followup_visit (0.76, CI 0.61-0.95), adjusted for the other factors, with the usual observational caveats
2 a model always looks better on data it trained on; only held-out data shows how it will do on new patients (and catches overfitting)
3 pick a random readmitted and a random non-readmitted patient: about 72% of the time the model scores the readmitted one higher; 0.50 is a coin flip
4 with few events, guessing 9% for everyone already has a low squared error; look at the relative improvement, AUC and calibration too
5 predicted risks match observed rates (20% predicted, about 20% happen); staff use the number itself to set priorities and budgets
6 p says the effect is probably not zero given 4,000 patients; AUC says it barely separates who comes back. Real, small, not predictive
7 a list of the top 10% holds about 27% of readmissions at 2.7x the base rate: a focused, but far from complete, outreach list
8 e.g. simulated data, no external or time-based validation, possible bias across groups, features not available at discharge time (leakage)